

C Programming + Terminal — Quick Notes

Beginner cheat sheet • Mac + Clang

1. Basic C Program

```
#include <stdio.h>
int main()
{
    printf("Name: Aaryan Shah");
    return 0;
}
```

Term	Small explanation
#include <stdio.h>	Includes Standard Input/Output. Needed for printf().
int main()	Main function; program execution starts here. int means it returns an integer.
{ }	Marks the beginning and end of the function's code.
printf()	Prints/displays output on the screen.
"..."	Text/string written inside quotes.
;	Ends a C statement.
return 0;	Returns 0 from main; commonly indicates successful completion.

2. C Code → Executable → Output

Source code (.c) → Compiler (Clang) → Executable → run → Output

1st.c → clang → 1st → ./1st → Output

3. Compile on Mac

clang 1st.c -o 1st

Part	Meaning
clang	C compiler.
1st.c	Source-code file.
-o	Output option: choose the executable name.
1st	Name of the executable.

4. Run

./1st

. = current directory; / = path separator; 1st = executable name.

5. Common Error

-0 ■ zero -o ■ small letter O

Correct: clang 1st.c -o 1st

6. Terminal Workflow

Step	Command	Purpose
1	ls	See files
2	clang 1st.c -o 1st	Compile
3	./1st	Run
4	Output	See program result

Quick memory: .c = source code • clang = compiler • -o = output name • ./ = run from current folder • main() = starting point